University of Engineering & Management, Kolkata
Institute of Engineering and Management, Newtown Campus
Department of Computer Applications

ASSIGNMENT SUBMISSION — TOP SHEET

Technical Class Assessment — April 2026
Database Management Systems & Java with JDBC | MCA Programme

STUDENT DETAILS

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Section	A	**Semester**	2
Programme	MCA	**Academic Year**	2025-26
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ASSIGNMENT DETAILS

Subject	Java-DBMS Assignment	**Set Number**	7
Problem Title	The Scholarship Portal		
Language Used	DBMS & Core Java	**Database Used**	MySQL 8.10.8
Submission Date	10/04/2026	**Total Pages**	13

SUBMISSION CHECKLIST

#	Item	Included (✓/✗)	Pages / Remarks
1	Printed Java Source Code (DBConnection.java)	✓	
2	Printed Java Source Code (Main.java)	✓	
3	Printed SQL Schema File (schema.sql)	✓	

4	Screenshot(s) of Program Execution (printed)	✓	
5	Problem Statement (this top sheet attached)	✓	

DECLARATION

I hereby declare that this assignment is my own original work. I have not copied from any other student or used any unauthorised sources. I understand that plagiarism will result in zero marks and disciplinary action.

10/04/2026

Student Signature & Date

Faculty Signature & Date

TECHNICAL CLASS ASSESSMENT — April – 2026

Degree: MCA	Stream: MCA
Due Date: _____	**Set No: 7**

The Scholarship Portal

[Domain: Education]

Language	Core Java (Console Application)
Database	MySQL
Connectivity	JDBC (Java Database Connectivity)

The Story

> *"A student with 82% and family income under 3 lakhs should get the scholarship automatically. Why does it take a month?"*

A state government scholarship office receives thousands of applications each year. Applications are filed on paper, eligibility is checked manually, and approved students are informed weeks late. Many deserving students miss the deadline simply because their file got lost.

Your task is to build a Scholarship Management System using Java and MySQL.

Your Task

Build a Scholarship Application System as a Java console application using JDBC and MySQL.

Your system must allow:

- Submit a new scholarship application (student name, percentage, annual income)
- Automatically set eligibility: marks >= 75 AND income <= 300000
- Approve an eligible application (update status to approved)
- Display all approved students and total scholarship amount (assume Rs 10,000 per student)

Database Schema

Create the following table(s) in MySQL:

```
CREATE TABLE applications (
    app_id      INT PRIMARY KEY AUTO_INCREMENT,
    student     VARCHAR(100),
    percentage  DECIMAL(5,2),
    income      INT,
    eligible    BOOLEAN,
    status      ENUM('pending','approved','rejected') DEFAULT 'pending',
```

```
    applied_on  DATE DEFAULT (CURRENT_DATE)
);
```

Required Features

Feature	Implementation Hint
Submit Application	INSERT; set eligible = (percentage>=75 AND income<=300000)
Approve Application	UPDATE status='approved' WHERE app_id=? AND eligible=TRUE
View Approved	SELECT * FROM applications WHERE status='approved'
Total Payout	SELECT COUNT(*) * 10000 WHERE status='approved'

Constraints

- Use Core Java only — no Spring, Hibernate, or web frameworks
- Connect to MySQL using JDBC (no JPA / ORM)
- All operations must be menu-driven via console
- No GUI required — Scanner-based input is sufficient
- Handle SQL exceptions with try-catch blocks
- Use PreparedStatement for all SQL queries

Sample Interaction

```
===== Scholarship Portal =====
1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Exit
Enter choice: 1
Student: Ananya Roy  Marks%: 88.5  Annual Income: 240000
Application submitted. Eligible: YES
```

Submission Checklist

- DBConnection.java (JDBC connection class)
- Main.java (menu-driven main program)
- SQL file: schema.sql (CREATE TABLE statements)
- screenshot.png (terminal output showing at least 2 features working)

> *“A database is not just data. It is the memory of your application. Make it reliable.”*

src/main.java:

```
import java.sql.*;
import java.util.Scanner;

public class Main {

    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        System.out.println("===== Scholarship Portal =====");

        while (true) {
            System.out.println("\n1. Submit Application");
            System.out.println("2. Approve Application");
            System.out.println("3. View Approved Students");
            System.out.println("4. Total Scholarship Payout");
            System.out.println("5. Total Applications Submitted");
            System.out.println("6. Exit");
            System.out.print("Enter choice: ");

            int choice = sc.nextInt();
            sc.nextLine();

            switch (choice) {
                case 1: submitApplication(); break;
                case 2: approveApplication(); break;
                case 3: viewApproved(); break;
                case 4: totalPayout(); break;
                case 5: totalApplications(); break;
                case 6:
                    System.out.println("Goodbye!");
                    System.exit(0);
                default:
                    System.out.println("Invalid choice. Try
again.");
            }
        }
    }

    static void submitApplication() {
        System.out.print("Student Name: ");
        String name = sc.nextLine();

        System.out.print("Marks% (e.g. 88.5): ");
        double percentage = sc.nextDouble();

        System.out.print("Annual Income: ");
        int income = sc.nextInt();

        boolean eligible = (percentage >= 75 && income <= 300000);
```

```
        String sql = "INSERT INTO applications (student, percentage,
income, eligible) VALUES (?, ?, ?, ?)";

        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, name);
            ps.setDouble(2, percentage);
            ps.setInt(3, income);
            ps.setBoolean(4, eligible);
            ps.executeUpdate();

            System.out.println("Application submitted ");

        } catch (SQLException e) {
            System.out.println("Error submitting application: " +
e.getMessage());
        }
    }

    static void approveApplication() {
        System.out.print("Enter Application ID to approve: ");
        int appId = sc.nextInt();

        String sql = "UPDATE applications SET status='approved'
WHERE app_id=? AND eligible=TRUE";

        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, appId);
            int rows = ps.executeUpdate();

            if (rows > 0) {
                System.out.println("Application " + appId + "
approved successfully.");
            } else {
                System.out.println("Student is not eligible.");
            }

        } catch (SQLException e) {
            System.out.println("Error approving application: " +
e.getMessage());
        }
    }

    static void viewApproved() {
        String sql = "SELECT * FROM applications WHERE
status='approved'";
```

```
        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql);
             ResultSet rs = ps.executeQuery()) {

            System.out.println("\n--- Approved Students ---");
            System.out.printf("%-5s %-20s %-10s %-12s%n", "ID",
"Name", "Marks%", "Income");
            System.out.println("-----------------------------------
---------");

            boolean found = false;
            while (rs.next()) {
                found = true;
                System.out.printf("%-5d %-20s %-10.2f %-12d%n",
                    rs.getInt("app_id"),
                    rs.getString("student"),
                    rs.getDouble("percentage"),
                    rs.getInt("income"));
            }

            if (!found) System.out.println("No approved students
yet.");

        } catch (SQLException e) {
            System.out.println("Error fetching data: " +
e.getMessage());
        }
    }

    static void totalPayout() {
        String sql = "SELECT COUNT(*) * 10000 AS total FROM
applications WHERE status='approved'";

        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql);
             ResultSet rs = ps.executeQuery()) {

            if (rs.next()) {
                System.out.println("Total Scholarship Payout: Rs. "
+ rs.getInt("total"));
            }

        } catch (SQLException e) {
            System.out.println("Error calculating payout: " +
e.getMessage());
        }
    }

    static void totalApplications() {
        String sql = "SELECT COUNT(*) AS total FROM applications";
```

```
        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql);
             ResultSet rs = ps.executeQuery()) {

            if (rs.next()) {
                System.out.println("Total Applications Submitted: "
+ rs.getInt("total"));
            }

        } catch (SQLException e) {
            System.out.println("Error fetching count: " +
e.getMessage());
        }
    }
}
```

src/DBConnection.java:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {

    // Change these to match your MySQL setup
    private static final String URL =
"jdbc:mysql://localhost:3306/scholarship_db";
    private static final String USER = "root";
    private static final String PASSWORD = "Kaustav@02#.me";

    public static Connection getConnection() throws SQLException {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            return DriverManager.getConnection(URL, USER, PASSWORD);
        } catch (ClassNotFoundException e) {
            throw new SQLException("MySQL JDBC Driver not found!",
e);
        }
    }
}
```

sql/schema.sql:

```
CREATE DATABASE IF NOT EXISTS scholarship_db;
USE scholarship_db;

CREATE TABLE IF NOT EXISTS applications (
    app_id      INT PRIMARY KEY AUTO_INCREMENT,
    student     VARCHAR(100),
    percentage  DECIMAL(5,2),
    income      INT,
    eligible    BOOLEAN,
    status      ENUM('pending', 'approved', 'rejected') DEFAULT
'pending',
    applied_on  DATE DEFAULT (CURRENT_DATE)
);
```

Screenshots:

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 4
Total Scholarship Payout: Rs. 40000
```

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 5
Total Applications Submitted: 23
```

app_id	student	percentage	income	eligible	status	applied_on
1	kingshuk ghosh	90.00	100000	1	approved	2026-03-31
2	Sayan Ghosh	56.00	96000	0	pending	2026-04-04
3	Deep Das	96.00	300000	1	pending	2026-04-04
4	Joy Sen	89.00	500000	0	pending	2026-04-04
5	Jhon Adhikari	96.00	250000	1	pending	2026-04-04
6	Kartik Ghoshal	83.00	630000	0	pending	2026-04-04
7	Sayak Banerjee	89.00	250000	1	pending	2026-04-04
8	Sudip Ghosh	80.00	260000	1	pending	2026-04-04
9	Joy Das	90.00	450000	0	pending	2026-04-04
10	Mita Sen	90.00	380000	0	pending	2026-04-04
11	Rishi DasGupta	89.00	280000	1	pending	2026-04-04
12	Rahi Sen	86.00	25000	1	approved	2026-04-04
13	Dipanjan Das	74.00	18000	0	pending	2026-04-04
14	Heera Mukherjee	87.00	200000	1	pending	2026-04-04
15	Sujal Sen	78.00	15000	1	pending	2026-04-04
16	Sahil Dey	89.00	270000	1	pending	2026-04-04
17	Sohini Choudh...	85.00	290000	1	approved	2026-04-04
18	Aditi Ghosh	96.00	85000	1	pending	2026-04-04
19	Joy Kumar	18.00	280000	0	pending	2026-04-04
20	Dibanjan Deb	98.00	290009	1	pending	2026-04-04
21	Heema Sen	96.00	280000	1	pending	2026-04-04
22	Garima Das	95.00	180000	1	approved	2026-04-04
23	Deep Sen	98.00	254000	1	pending	2026-04-04
NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 3

--- Approved Students ---
ID    Name                Marks%     Income
-----------------------------------------
1     kingshuk ghosh      90.00      100000
12    Rahi Sen            86.00      25000
17    Sohini Choudhury    85.00      290000
22    Garima Das          95.00      180000
```

```
1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 1
Student Name: Deep Sen
Marks% (e.g. 88.5): 98
Annual Income: 254000
Application submitted
```

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 1
Student Name: Garima Das
Marks% (e.g. 88.5): 95
Annual Income: 180000
Application submitted
```

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 2
Enter Application ID to approve: 10
Student is not eligible.
```

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 2
Enter Application ID to approve: 22
Application 22 approved successfully.
```

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 2
Enter Application ID to approve: 17
Application 17 approved successfully.
```

```
===== Scholarship Portal =====

1. Submit Application
2. Approve Application
3. View Approved Students
4. Total Scholarship Payout
5. Total Applications Submitted
6. Exit
Enter choice: 2
Enter Application ID to approve: 12
Application 12 approved successfully.
```